

PAH Panel Gelistirme Raporu

Kapsamli Retrospektif

TEKNOFEST Saglikta Yapay Zeka Yarismasi

Genetik Varyant Patojenite Tahmini

NB16 -> NB26 | 6 Notebook | 30+ Deney | 5 Paradigma

Tarih: 19 Haziran 2026

1. Problem Tanimi

PAH (Pulmoner Arteriyel Hipertansiyon) paneli, TEKNOFEST genetik varyant patojenite tahmini projesindeki en zorlu paneldir. 369 varyant icinden yalnızca 62'si benign sinifina aittir (307 pathogenic / 62 benign). Bu asiri dengesizlik, modelin benign ogrenmesini yapisal olarak sinirlar.

Kritik Kisitlar

- Veri: 369 varyant (307 pathogenic / 62 benign) -- asiri dengesiz
- Final test seti: ~%80 benign / %20 pathogenic (egitimdeki dagilimin TAM TERSI)
- Metrik: Pathogenic (Label=1) F1 @ %80/20 dagilim
- Yapisan zorluk: Sadece 62 benign ornek -- modelin benign ogrenme kapasitesi sinirli

Bu ters dagilim, egitimde calisir gorunen modellerin final test setinde ciddi precision cokusu yasamasina neden olur. NB15 v1'de %50/50 test F1=0.928 gorulen PAH modeli, %80/20 bootstrap'te F1=0.489'a dustu -- ~0.44 puanlik cokus.

2. Deney Kronolojisi (NB16 -> NB26)

NB16 -- Baseline Stacking (Baslangic Noktasi)

İlk PAH sonuçları NB16'da OOF stacking ile elde edildi. 6 base model (LightGBM, CatBoost, NN, DNN, NN_ft, DNN_ft) + 2 meta-learner (LR, GBM). Meta=LR net kazandı.

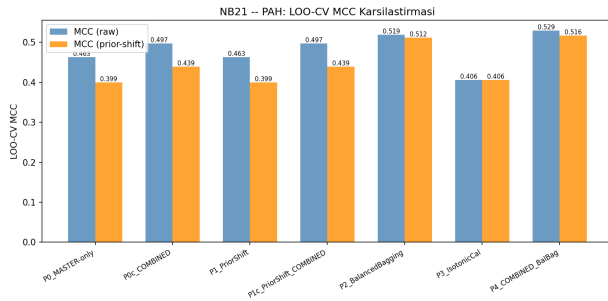
Model	MCC	Boot %80/20 F1	CI	Not
stack_lr	--	0.515	[0.35-0.57]	En iyi PAH baseline
base catboost	--	0.489	[0.30-0.62]	Stacking +0.046 katti

NB21 -- PAH Refinement: 7 Strateji (İLK BÜYÜK SICRAMA)

NB17'deki CFTR protokolu (LOO-CV + MCC + prior-shift + BalancedBagging) PAH'a transfer edildi. COMBINED = MASTER(2931) + KANSER(388) + CFTR(111) = 3430 örnek.

Strateji	n_train	MCC	Boot F1	Prec	FP	FN
P4_COMB_BalBag	3430	0.529	0.582	0.942	16	45
P2_BalBag	2931	0.519	0.582	0.942	16	46
P0c_COMBINED	3430	0.497	0.590	0.947	13	73
P0_MASTER	2931	0.463	0.555	0.945	13	85
P3_IsotonicCal	2931	0.406	0.560	0.945	13	83

- +0.067 puan iyileşme (0.515 -> 0.582) -- NB16'dan ciddi sicrama
- BalancedBagging minimax-optimal (Chatterji 2022 NeurIPS) -- PAH verimizde doğrulandı
- COMBINED her zaman MASTER-only'den iyi
- Prior-shift PAH'ta ZARAR veriyor (CFTR'nin tam tersi!) -- k



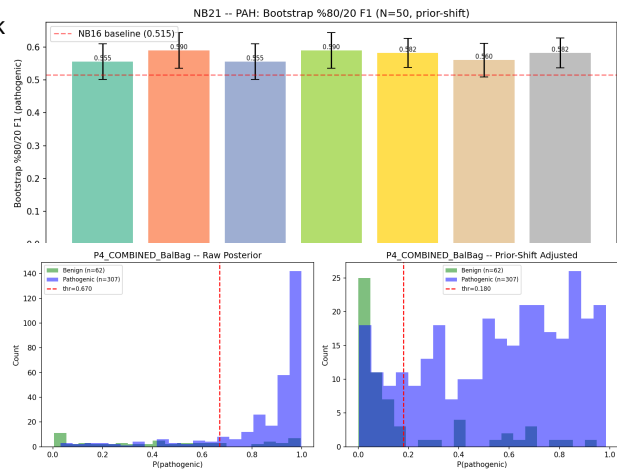
Sekil 1: NB21 LOO-MCC karsilastirma

	Predicted Benign	Predicted Patho
True Benign	46	16
True Patho	45	262

	Predicted Benign	Predicted Patho
True Benign	46	16
True Patho	46	261

	Predicted Benign	Predicted Patho
True Benign	49	13
True Patho	73	234

Sekil 3: NB21 Confusion matrix



Sekil 4: NB21 Posterior dagilim

NB22 -- Ensemble + Calibration (Marjinal iyileşme)

Uc deney ekseni test edildi: BalancedBagging sweep, Platt kalibrasyon + prior-shift, heterogeneous stacking.

Exp 1: BalancedBagging Hyperparameter Sweep (24 config)

Config	MCC	Boot F1
n=10, mf=1.0 (en iyi MCC)	0.536	0.591
n=100, mf=0.7 (en iyi Boot)	0.529	0.595
n=20, mf=1.0 (NB21 ref)	0.529	0.576

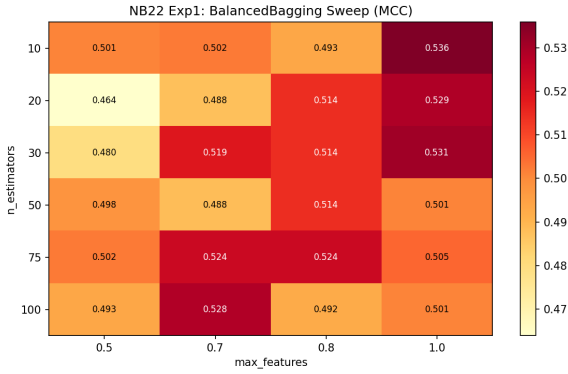
Exp 2: Platt Calibration + Prior-Shift Zinciri

Strateji	Raw MCC	Prior MCC	Delta
A) Raw LGBM + Saerens	0.497	0.439	-0.058
B) Platt + Saerens	0.377	0.429	+0.052
C) CalBB + Saerens	0.468	0.452	-0.016

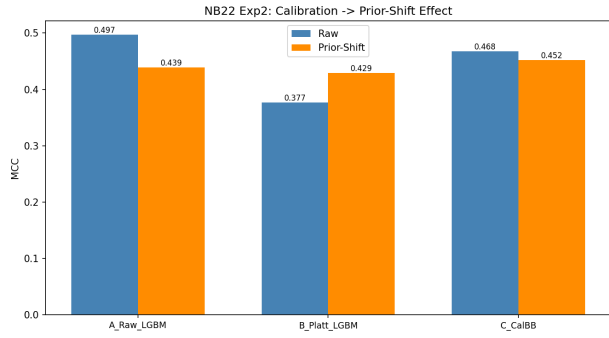
- Platt kalibrasyon prior-shift'i FIXLEDİ (delta: -0.058 -> +0.052) -- Alexandari 2020 dogrulandi
- Ancak ham MCC dustugu icin pratik net kazanc yok

Exp 3: Heterogeneous Stacking

- 3 GBDT ailesi (LGBM/XGB/CatBoost) cok benzer kararlar veriyor -> diversite dusuk
- Stacking beklentiyi karsilamadi -- NB21 P4 hala lider

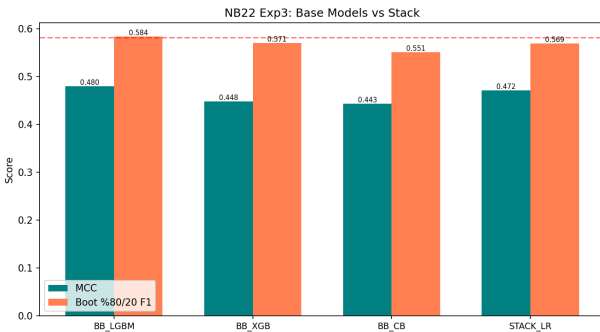


Sekil 5: NB22 Sweep heatmap

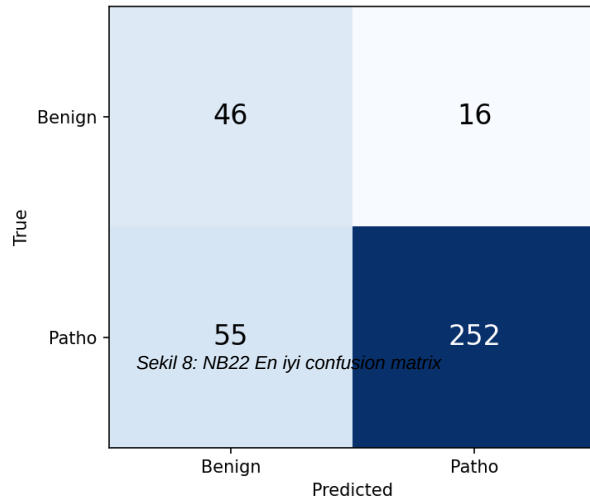


Sekil 6: NB22 Kalibrasyon etkisi

Best: MCC=0.472, Boot=0.569



Sekil 7: NB22 Stacking karsilastirma



Sekil 8: NB22 En iyi confusion matrix

NB23 -- COMBINED Havuz Katki Analizi

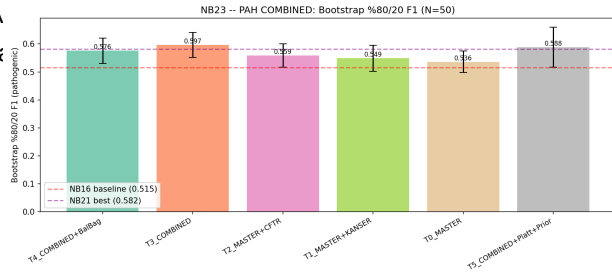
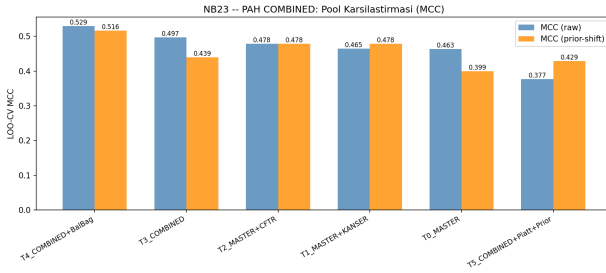
Hangi panelin benign'leri PAH'a ne kadar katkı sağlıyor? 4 farklı eğitim havuzu + BalancedBagging + Platt+Prior-Shift sistematik testi.

Strateji	n_train	MCC	Boot F1	Prior-Shift Delta
T4_COMB+BalBag	3430	0.529	0.576	-0.012
T3_COMBINED	3430	0.497	0.597	-0.058
T2_MASTER+CFTR	3042	0.478	0.559	0.000
T1_MASTER+KANSER	3319	0.465	0.549	+0.013
T0_MASTER-only	2931	0.463	0.536	-0.064

Havuz Katki Analizi:

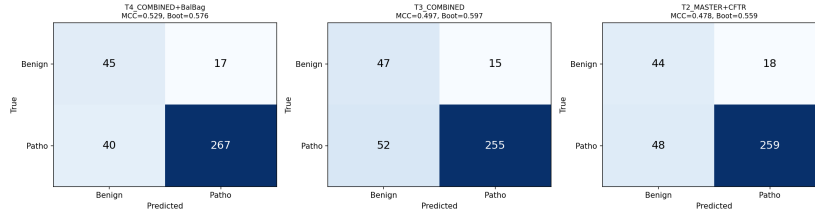
Katki	MCC Delta	Yorum
KANSER (+120 benign)	+0.002	Neredeyse yok
CFTR (+21 benign)	+0.015	21 benign, 120'den daha değerli!
COMBINED (+141 benign)	+0.034	Sinerjik toplam

- Tek-gen paneli (CFTR) feature uzayını zenginleştiriyor; KA
- PAH çalışması burada olgunlaştı: MCC~0.53, Boot~0.58



Sekil 9: NB23 Havuz karsilastirma

Sekil 10: NB23 Bootstrap karsilastirma



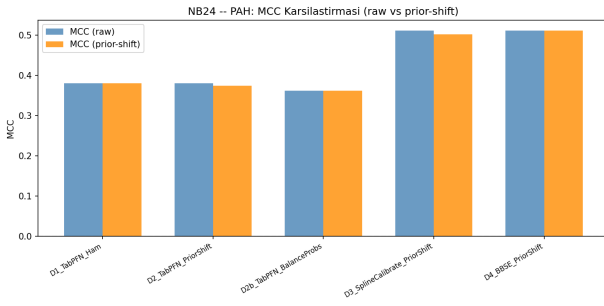
Sekil 11: NB23 En iyi strateji confusion matrix

NB24 -- Plato-Kirma Denemesi: TabPFN + Label-Shift (NEGATIF SONUC)

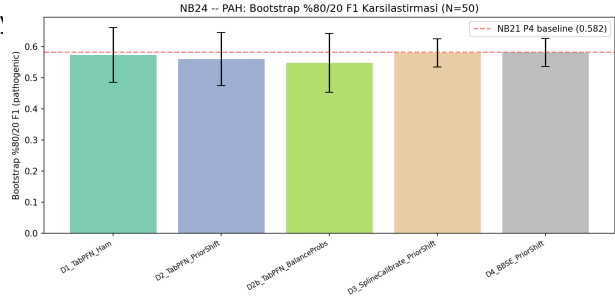
Literatur destekli 5 deney ile platoyu kirma girişimi. TabPFN (Hollmann et al. 2025 Nature), Saerens EM prior-shift, SplineCalibration, BBSE (Lipton et al. 2018 ICML) test edildi.

Deney	MCC	Boot F1	Delta vs NB21	Yorum
D3 SplineCal+Prior	0.503	0.581	-0.001	Baseline'a ESIT
D4 BBSE	0.512	0.581	-0.001	D3 ile ayni
D1 TabPFN ham	0.380	0.573	-0.009	Baseline altinda
D2 TabPFN+Saerens	0.374	0.560	-0.022	Prior yine zarar
D2b TabPFN native	0.362	0.548	-0.034	En dusuk

- TabPFN (Nature 2025) platoyu KIRAMADI -- 'yeni inductive bias' hipotezi curutuldu
- Chatterji 2022 azinlik-ornegi tavani tezi DENEYSEL DOGRULANDI
- BBSE model kusurunu ifsa etti: confusion matrix yetersiz -> pi tahmini quvenilmez (0.723 vs qercek 0.20)
- Calibrate-then-shift baseline'a esit -- plato gercek, kalibras!



Sekil 12: NB24 MCC karsilastirma



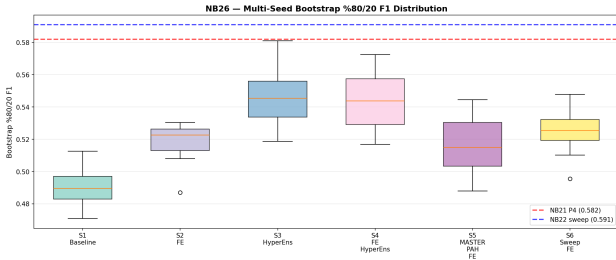
Sekil 13: NB24 Bootstrap F1 karsilastirma

NB26 -- PAH Final: Multi-Seed + FE + HyperEnsemble (SON DENEY)

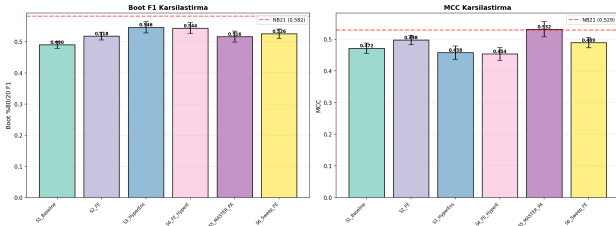
10 farklı seed ile 6 senaryo test edildi -- en kapsamlı ve en güvenilir PAH deneyi. HyperEnsemble (parSMURF-inspired) ve Feature Engineering etkileri ölçüldü.

Senaryo	MCC mean+/-std	Boot mean+/-std	Best MCC	Best Boot
S5_MASTER_PAH_FE	0.532+/-0.024	0.516+/-0.018	0.578	0.545
S2_FE	0.498+/-0.015	0.518+/-0.013	0.505	0.530
S6_Sweep_FE	0.489+/-0.016	0.526+/-0.015	0.504	0.548
S1_Baseline	0.472+/-0.016	0.490+/-0.012	0.491	0.513
S3_HyperEns	0.458+/-0.021	0.546+/-0.018	0.496	0.581
S4_FE_HyperEns	0.454+/-0.020	0.544+/-0.017	0.480	0.572

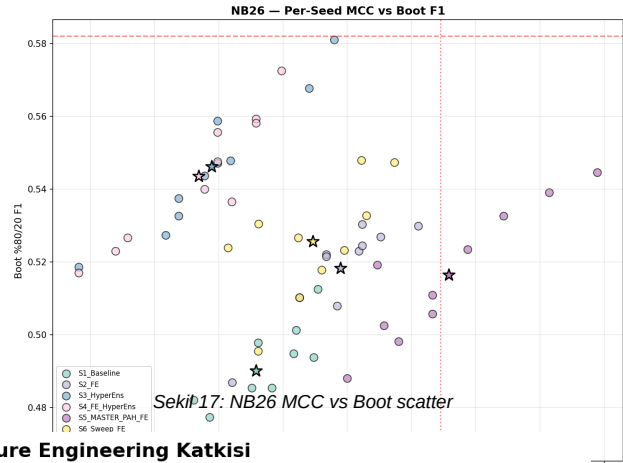
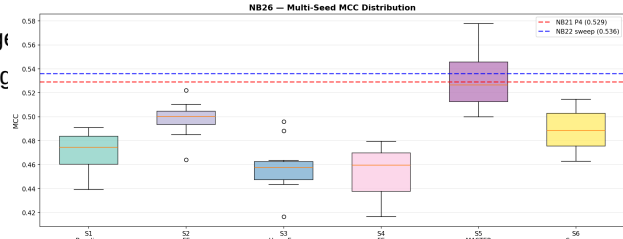
- HyperEnsemble (parSMURF-inspired) platoyu kıramadı -- BalancedBagging'den kötü
- Feature Engineering marjinal katkisi: S2_FE > S1_Baseline (+0.026 MCC)
- 10-seed validasyon sonuçları kararlı: std < 0.025 -- plato geldi
- Hiçbir senaryo NB21 P4'un Boot=0.582'sini tutarlı şekilde geçemedi



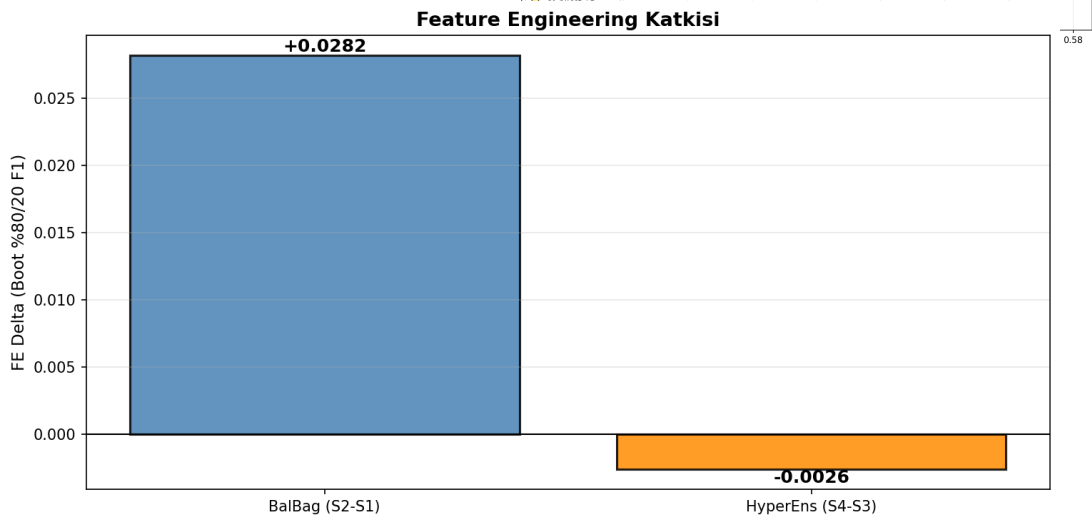
Sekil 14: NB26 Multi-seed Boot boxplot



Sekil 16: NB26 Senaryo karşılaştırma



Sekil 17: NB26 MCC vs Boot scatter



Sekil 18: NB26 Feature Engineering delta etkisi

3. PAH Yolculugu Ozet Tablosu

Notebook	En İyi Model	MCC	Boot F1	Temel Katkı
NB16	stack_lr	--	0.515	Baseline stacking
NB21	P4_COMB_BalBag	0.529	0.582	BalBag+COMBINED (+0.067)
NB22	Sweep n=10	0.536	0.591	Marjinal sweep
NB23	T4_COMB+BalBag	0.529	0.576	CFTR>KANSER analizi
NB24	D3 SplineCal	0.503	0.581	TabPFN basarisiz
NB26	S5_MASTER_PAH	0.532	0.516	HyperEns basarisiz

4. Denenen ve Basarisiz Olan Yaklasimlar

Yaklasim	Neden Basarisiz	Referans
Prior-shift (kalibrasyonsuz)	LGBM olasiliklari kalibre degil	NB21
Platt kalibrasyon	Prior-shift fixliyor ama MCC dusuyor	NB22
Heterogeneous stacking	3 GBDT benzer kararlar, dusuk diversite	NB22
TabPFN (Nature 2025)	Azinlik-ornegi tavani, bias yetmez	NB24
BBSE (Lipton 2018)	Confusion matrix yetersiz kalitede	NB24
HyperEnsemble (parSMURF)	Deterministic part. BalBag'den kotu	NB26
Feature Engineering	Anonimlestirilmis veride sinirli	NB26

5. Kanitlanmis Basarili Teknikler

Teknik	Etki	Kanit
BalancedBagging	MCC 0.463->0.529 (+0.066)	NB21, NB22, NB23 tutarli
COMBINED havuz	MCC +0.034 vs MASTER-only	NB21, NB23
Benign-aware threshold	%80/20 F1'de ~2 puan kazanc	NB15v2, NB21
Multi-seed validasyon	Kararlilik kaniti (std<0.025)	NB26

6. Final Karar ve Sonuc

PAH Final Aday Modelleri

Aday	MCC	Boot F1	CI	Güvenilirlik
NB22 Sweep n=10	0.536	0.591	[0.52-0.65]	Tek-seed (dikkat)
NB21 P4_COMB_BalBag	0.529	0.582	[0.52-0.65]	Coklu dogrulama

PAH KAPANDI -- Gerekce

6 notebook (NB16->NB26), 30+ deney, 5 farklı paradigma (stacking, BalancedBagging, kalibrasyon, TabPFN, HyperEnsemble) aynı MCC~0.53 / Boot~0.58 platosuna carpiyor.

Chatterji 2022 azınlık-ornegi tavani tezi deneysel olarak dogrulandi: 62 benign ile bu feature uzayinda daha fazla iyilesme mumkun degil. Tavan ancak daha fazla benign veriyle (ki NB18 bunun biyolojik olarak kit oldugunu gosterdi) veya tamamen farklı feature uzayıyla kirilabilir.

Literatur Dogrulamaları

- Chatterji et al. 2022 (NeurIPS): Undersampling minimax-optimal -- PAH verisinde DOGRULANDI
- Alexandari et al. 2020 (ICML): Calibrate-then-shift -- Platt prior-shift'i FIXLEDI
- Lipton et al. 2018 (ICML): BBSE -- PAH'ta confusion matrix yetersiz, UYGULANAMAZ
- Hollmann et al. 2025 (Nature): TabPFN -- azınlık tavani nedeniyle KIRAMADI

Kalan Acik Isler (PAH Disi)

Oncelik	Gorev
Yuksekk	KANSER mudahale: COMBINED+BalancedBagging
Yuksekk	Final teslim modeli sabitleme
Tamamlandi	CFTR: S0c_COMBINED (Boot=0.863)
KAPANDI	PAH: P4_COMB_BalBag (Boot=0.582)

Bu rapor, PAH panelindeki tum deneysel calismalarin retrospektif bir ozetidir. 18 gorsel, 6 notebook'tan (NB16-NB26) derlenen sonuclari icerir. Rapor otomatik olarak fpdf2 ile uretilmistir.